

Predicting Financial Crisis through emotion explosiveness in news articles

Rohan Bajaj: AE29565

Supervisor: Filippas Papakonstantinou



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Declaration

I, Rohan Bajaj, a student at King's College London, hereby declare that this dissertation, titled “Predicting Financial Crisis through Emotion Explosiveness in News Articles”, is a result of my own independent research and original work. This dissertation is submitted as a requirement for the completion of my degree, MSc in Finance Analytics. I have carried out this research with full adherence to the academic and ethical standards as required by King's College London.

All sources of information, data, and scholarly insights that have contributed to the development of this research have been properly acknowledged and referenced in accordance with the university's guidelines on academic integrity and plagiarism. Wherever necessary, I have sought to cite the work of others with the utmost precision, and any external material used is explicitly credited.

Furthermore, I confirm that this dissertation has not been submitted, either in full or in part, for any other academic degree or professional qualification at King's College London or any other institution.

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Rohan Bajaj

Abstract

The research focusses on capturing human emotions like trust, fear, sadness, etc., from various news articles, and quantifying such emotions to predict a possible upcoming financial crisis.

The dictionary used for the model has been created by first using a base lexicon from Mohammad, S. M., & Turney, P. D. (2013) referred to as the (National Research Council Canada) NRC Lexica. This was further expanded by pairing the terms with relevant financial, economic, business and political terms extracted by running Latent Dirichlet Allocation on the Guardian articles.

The first stage of the model placed these self-constructed dictionaries alongside each article and calculated a similarity score using methods such as Levenshtien ratio, Cosine similarity, Jaccard similarity and used word embeddings to find Euclidean distances.

The second stage of the model used Principal Component analysis to extract the key Principal Component for each emotion, calculated using the scores for each of these emotions from all the similarity methods used, separately. Thus, arriving at the Principal Components which further aided in reducing the dimensionality of the problem. These Principal Components were then put through a Multilinear Regression which generated a statistic that provided signals for upcoming crisis.

Section 1:
Introduction

This section begins with a background on text analytics and sentiment analysis and is followed by a description of the concept of “big data”. The purpose of this section is also to appraise the reader about various facets of this study including Objective, Scope of the study, hypothesis, rationale and literature review.

1.1 - Background

Text analytics is a broad field which comes under the scope of Data Science and Artificial Intelligence that focusses on extracting necessary signals and meaningful information from unstructured textual data. This involves performing a wide range of techniques such as Natural Language Processing (NLP), information retrieval and data mining through web-scraping and APIs (Application Programming Interface). The primary goal of text analytics is to convert readily available textual data into a structured format which can further be used to analyze patterns, signals and trends.

The inputs used within text analytics come from raw text which is produced by both: individuals, such as: Twitter and Facebook comments and posts and organizations, such as: press releases, market outlook commentary, podcast transcriptions and earnings reports. As the volume of this data has grown over time, its importance in educational research and industrial application has also increased significantly. Organizations are currently using text analytics to analyse customer feedback, social media comments, and internal documents to improve decision-making and strategic planning.

Sentiment analysis (also referred to as opinion mining) is an application of text analytics which aims to determine the tone expressed in the given textual data. It categorizes text as positive, negative, or neutral, and can also be used to delve deeper into specific emotions such as joy, sadness and anger. This technique has become an important tool in extracting public opinion to be able to predict market trends or customer opinions. Sentiment analysis has a wide scope of applicability and is being used in various fields, including marketing, finance and politics.

The methodology of sentiment analysis involves a multitude of steps before reaching the final step of opinion mining. The raw data needs to be pre-processed to achieve a desirable structure for the sentiment analysis techniques. Pre-processing focuses on cleaning the data and removing words and phrases which are not indicative of any human emotion or sentiment, often referred to as stop-words such as: “a”, “and”, “the”, “is”, “are” and so on. This may also include punctuation marks and any numerical data within the text depending on the working of the textual model. Once the data is pre-processed, it is then put through a model which helps generate results about the sentiment of the document.¹

“Big-Data” refers to a dataset characterized by a large number of independent variables, attributes and a wide distribution structure which increases the complexity of analysis. With the rise in volumes, it becomes challenging to detect the key signals which are necessary for opinion mining. Multiple pre-existing techniques tend to fail in such scenarios or are unable to retain accuracy in the presence of such high volumes of data. Thus, to tackle this problem, this high dimensional data is represented using a smaller sub-space retaining an optimum number of characteristics to ensure

¹ <https://careerfoundry.com/en/blog/data-analytics/text-analysis/#:~:text=Text%20analysis%20is%20a%20powerful%20machine-learning%20technique%20used,topics%2C%20and%20measure%20customer%20sentiment%2C%20among%20other%20things.>

appropriate interpretability. This includes techniques such as Principal Component Analysis, penalized regressions or subset selection techniques.²

1.2 - Objective

The primary objective of this study is to conduct an in-depth analysis of the potential of leveraging emotional sentiment data, particularly emotions prevailing among the investor groups as a forecasting tool in predicting an upcoming financial crisis.

The study also explores whether these selected emotions, when analysed and quantified, can serve as a reliable indicator of market behaviour and aims to examine the effects of combining this sentiment analysis with the PSY test—a leading indicator in predicting bubbles in prices to improve the early detection and prediction of potential financial bubbles.

By integrating emotion and sentiment data with traditional financial indicators, this study seeks to develop a more nuanced and comprehensive model to forecast market movements and identify early warning signs of market instability, thereby contributing to the field of financial forecasting, text analytics and risk management.

1.3 - Scope

This paper aims to explore the potential of using sentiment analysis from an emotion specific perspective, instead of a positive, negative or neutral classification, as a predictive tool for stock market indices, with major focus on understanding how specific terms and phrases present in financial literature can indicate specific emotions which can further help in deciphering market trends on a given day. The primary focus of this study is on analyzing sentiment data derived from newspaper articles and pairing this with the PSY test, a leading indicator for time-series explosiveness, to predict market bubbles well in advance.

The study uses a major stock index of the UK markets: FTSE 100, covering a time frame from January 2000 to December 2021 and newspaper articles published by “The Guardian” for the same period. Data is sourced from the free to use publicly available APIs by guardian and Yahoo Finance and the research methodology involves quantitative analysis using similarity algorithms, Principal Component Analysis and Multilinear Regressions to correlate sentiment data with market movements.

However, this research will not extend to analysis of sentiment data from non-English speaking regions or niche financial markets. Additionally, it will not account for macroeconomic variables outside of those captured through the textual data. Limitations include a potential bias in the published articles and challenges of accurately quantifying each emotion

1.4 - Hypothesis

Emotional level, primarily amongst retail investors significantly influences stock market behavior leading to an observable fluctuation within the market indices.

² <https://journalofbigdata.springeropen.com/articles/10.1186/s40537-020-00320-x#:~:text=The%20simplest%20approach%20to%20handle,the%20problem%20of%20high%20dimensionality>

The rise in retail investor participation has led to a higher proportion of trades executed due to emotional responses and media influence, which correlates with the increased market volatility and trading volumes and is evidenced by trends in the equity markets.

Emotion specific signals present in financial literature can be captured and modelled to predict significant price movements which occur during crisis periods in market indices well ahead in time before the prices actually fall.

1.5 - Rationale

The rationale behind this study stems from the observation that retail investors are driven by emotions, sentiment and media rather than structured financial analysis. Their contribution to the equity market exchange trading volumes has increased significantly and trend has been notably seen within the UK markets, where retail participation surged by roughly 80% from an estimated 9.6% in 2015 to 16.5% in 2022 and rising, according to a Reuters report³. Parallely, an increase in total trading volumes across UK equity markets from £600 billion to £760 billion i.e. a growth of 23% over the same period. Retail participation within this increased from £55.2 billion to over £125 billion thus contributing to half the increase in volumes according to a London Stock exchange report and statistics.⁴

Given the role played by emotions in determining market outcomes, it is essential to understand and leverage these factors in predicting market movements. By analyzing the emotional sentiment prevalent within financial literature, this study seeks to explore the potential for developing predictive models that could provide valuable insights into future market trends.

1.6 - Literature Review

The research done so far in the field of text analytics under finance appears to have primarily focused on extracting sentiments from the existing text, and then performing a binary classification of whether the same expresses a positive or a negative sentiment. Such sentiments have been then used to determine the future direction that the equity markets may take.

Michelle Alexopoulos & John Cohen (2015) in their publication titled “The power of print: Uncertainty shocks, markets, and the economy” for example, explain how Wall Street behavior confirms the need for media data extraction, thus making text an important asset in the investment toolkit. They aimed to identify fluctuations in the stock market and based these fluctuations on central bank policies. Thus, to model this, Michelle and John created two main sets of dictionaries, and three subsets within each of those dictionaries.

They classified their input variable as being policy specific or general, and based on the flexibility of words within the dictionaries, there were three sub-categories under both variables: basic economic & basic uncertainties, basic economic & extended uncertainties and extended economic as well as uncertainties. The basic dictionaries consisted of the core terms while the extended versions of the dictionaries carried similar meanings but with some flexibility in their use. The precondition applied for using the extended versions was, presence of at least 1.5% of the basic terms in the text.

³ <https://www.reuters.com/finance>

⁴ <https://www.londonstockexchange.com/reports-and-statistics>

However, the nature of variables used to model the data seemed very restrictive and does not capture all the signals from the data that impact the price movements of stocks. The approach intended to model policy impact using the New York Times, which seemed like an inappropriate dataset for the variable that was attempted to be modelled. Instead, interbank communications or central bank reports would have been more appropriate for the intended research.

Moniz, A. and F. de Jong (2014) in their publication, "Predicting the Impact of Central Bank Communications" used central bank communications to perform price discovery. This seemed to cover the gaps of Michelle Alexopoulos & John Cohen in my opinion. Moniz, A. and F. de Jong modelled the data using a text rank link analysis algorithm. The idea behind their model was that central banks, after 2008 conveyed crucial information through their reports to allow investors to perform appropriate risk analysis. Since the goal of the central bank was to avert an upcoming crisis, the reports would contain essential information regarding upcoming bubbles and or crisis in the prices.

They found economic references and classified them into documents using a naïve bayes classifier. To measure the sentiment of these economic aspects, they used the term frequency of these aspects against the General Inquirer Dictionary and then employed link analysis approach, using the first two pages of Wikipedia for the keywords: 'inflation' & 'central bank communication'. The procedure involved finding frequently used words to perform topic discovery and identify the general themes across documents. These were then plotted (word vectorization), a text ranking algorithm was used to find the weight of each vertex to understand its importance. This was represented graphically in their report such that the size of the vertex corresponded with its rank (directly proportional). The computation of polarity was done by looking at the positive and negative words through the General Inquirer Dictionary (GIQ) to understand the tone with respect to prices, interest rates and growth rates.

The Central Bank comes out with its report 2 weeks after interest rate announcement which is every Thursday of the month. Moniz, A. and F. de Jong believed that investors await this report to take investment decisions and use Bank of England's investor estimates data of 12 months ahead Interest rate future forecasts, since in their research it was determined that this forecast provides for the most accurate predictions compared to 6-month or 3-month forecast. Thus, the model analyzed volume change for Interest rate 12-month future contracts between T-1 day of releasing the report and closing of after the report was released. Based on MSE (Mean Squared Error), RMSE (Root Mean Squared Error) and spearman's Rho a joint aspect polarity model was used, which combined the naïve tone with specific economic aspects like growth, inflation and interest rates and the directional LDA model, which had one direction for each of the 8 economic aspects.

The major downside of this model was the GIQ dictionary that was used, since it was not very specific to finance, there existed a high probability of overlooking key signals and capturing more noise while presenting the results. Additionally, given that the Central Bank report is released fortnightly, there is a very high probability of missing a signal through such a model due to low frequency of input data.

Samuel P. Fraiberger (2016) in their publication, "Media Sentiment and International Asset Prices" attempted to use Loughran and McDonald 2011a (L&M) dictionary in understanding economic sentiment via Reuters articles covering 12 countries from 1987 to 2013. Samuel's main aim was to use available economic indicators alongside the recently developed text analytics techniques to improve the pre-existing methods of estimating the GDP. This was done through the simplest bag-of-words approach, which was put alongside the Loughran and McDonald dictionary to check for

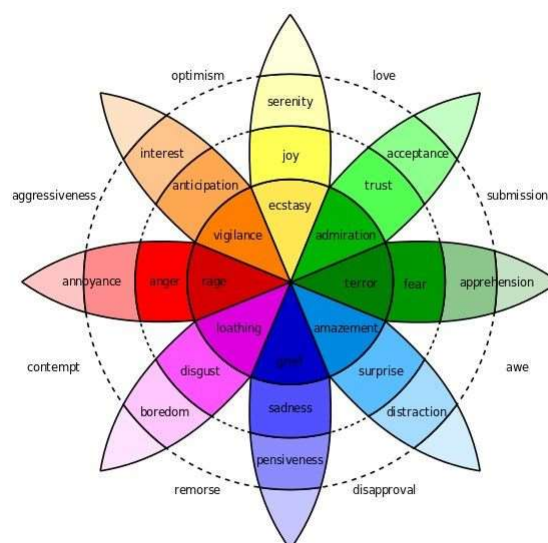
similarity on a binary level between positive and negative words. The key idea behind his research was that fluctuations in beliefs, which may be unrelated to the fundamental aspects of the equity market, also tend to impact macroeconomic variables and asset prices, which come from the waves of positive and negative sentiments in the market, as described by Keynes, J. M. (1936) in his book titled “animal spirits”.

Samuel’s research involved running predictive regressions of GDP on various forecasting variables and then adding his self-designed new sentiment index to improve these predictions. Overall, he could see a 19% reduction in the total forecast errors. This was done at different time horizons, which gave him the conclusion, that text techniques tend to work better over longer time horizons.

The Loughran and McDonald_dictionary used by Samuel P. Fraiberger comes from Loughran, T. and B. McDonald (2011b). A need for a field specific dictionary arose, when Loughran, T. and B. McDonald realized that the Harvard dictionary developed for the field of psychology was too general, and over 75% of the negative words within it were a part of neutral financial text and did not indicate any specific direction.

A vast majority of the dictionary-based research done focusses on the use of a pre-existing dictionary such as Harvard IV, GIQ or AFINN (Affective Norms for English Words). Harvard and GIQ consist of general terms and are more likely to overlook multiple sentiment signaling terms in the field of finance. The AFINN, though is predominantly characterized by financial terminologies, is only capable of performing a binary classification into a positive or a negative sentiment which narrows down the flexibility of the model workings and data interpretability.

Saif Mohammad and Peter Turney (2013) in their paper titled, “Crowdsourcing a Word Emotion Association Lexicon” emphasise on human reactions and how their scope extends beyond mere positive or negative reactions and aim to analyse various emotions through customer review data to understand an individual’s opinion about a particular product, company or policy. His research utilises the emotion analysis performed by Plutchik (1980). The psychological study stated eight primary emotions which form the foundation for all the other emotions. Plutchik proposed a wheel of emotions, where neighbouring emotions are expected to be similar to one another and opposite spokes indicate opposite emotions. Moreover, the colour of the wheel changes with intensity, such as anger being the strongest is displayed in red.

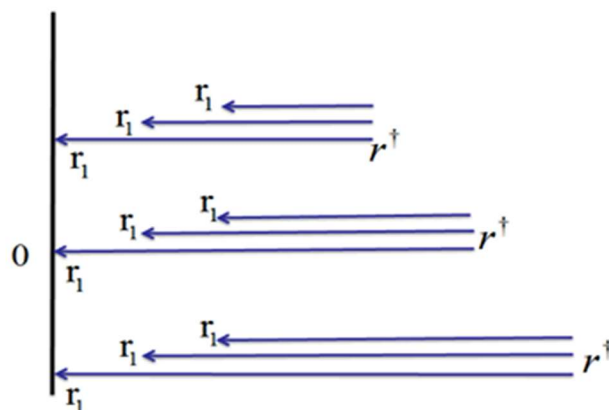


From Plutchik (1980)

There are two key facts that can be derived from the above-mentioned studies, firstly something more flexible than mere positive/negative classification is needed to provide higher flexibility in the model, and secondly a dictionary capable of modelling this flexibility. Thus, inspired by [Saif Mohammad and Peter Turney \(2013\)](#) and [Plutchik \(1980\)](#), an emotion-based dictionary has been created, capturing the 8 key emotions along with their relevant terminologies present in the financial literature to be able to predict the direction and magnitude of stock prices well in advance, depending on the combination and strength of emotions prevalent in the media.

This analysis is done using [Levenshtien \(1966\)](#) algorithm and other such methods for similarity with the emotional dictionary and using unsupervised techniques to discover statistics which would help predicting prices as a function of these emotions. The key basis behind using this method is that prices are always a function of demand and supply. Within the stock markets, investors tend not to follow a rational pattern during extreme phases and get carried away by emotion, thus leading to highly inflated or extremely bottomed out price levels which are different from their true fundamental values.

To validate the accuracy of the model, these similarity scores have been used in comparison with the ADF recursive algorithm developed by [Peter C. B. Phillips and Shuping Shi \(2018\)](#) in their paper titled “Testing for Multiple Bubbles: Historical Episodes of Exuberance and Collapse in the S&P 500” to identify bubbles or crisis. This technique uses a modified version of the traditional ADF T-statistic, whereby the user fixes an end point, and the algorithm would appropriately run recursively to find the most optimal start point as depicted in the diagram below:



from Philips Shuping and Jun Yu(2018)

Since emotion extraction would help predict the direction of upcoming swings in stock returns, it would imply that a period of extreme emotions would cause explosive behavior. This can be verified using the above-mentioned test for non-stationarity to ensure accuracy in predictions.

Section 2:
Data Composition

This section discusses the data structure which consists of three forms of data that were used to train the model to make appropriate predictions.

Firstly, the Guardian articles, based on which the model was trained to predict the prices. In total roughly 800,000 articles with a word count of over 100 million terms were used to perform the said analysis.

Secondly, the dictionary, which was formulated based on the Guardian articles downloaded through the API, was used as a measurement tool to investigate and measure the presence of emotions within the articles.

And lastly the dependent variable, which in this case was chosen as the FTSE 100 index to reflect the general price level of the UK equity markets.

2.1 - The Guardian Articles

Textual data was collected from Guardian, a popular UK based news reporting brand which provides a free developer API that was used to call for articles on their website for each date for a total period of twenty years. The period was limited to twenty years due to the limitations in data processing and modelling capabilities of the systems used.

The API allows to call for 200 articles per day, with a cap on the maximum number of calls to 500 for every 24 hours. Thus, restricting the total data download capacity to 500 days of articles daily. It was used to call for articles of five main section IDs namely: 'money', world, 'business', 'uk-news' and 'politics'.

2.1.1 - Money

This category includes articles related to personal finance, such as mortgage, loans, pensions, retirement planning etc. It includes various tips on managing debt, savings, budgeting strategies besides articles related to consumer rights.

The section on 'Money' even covers some broader aspects of the economy (new tax policies, changes in regulations, economic trends in the housing market, general price levels and interest rates) which impact individuals and their personal finances. There is also information about new insurance products, upcoming policies and general price levels for various kinds of premiums.



Raw word cloud for "Money" articles

As the name suggests, this section covers news from across the globe, providing updates on significant events in the form of investigative reports and breaking news.

world support back another attack family even case may told leave last week going decision kill want time found around year one Iraq new know last year many called country people law country people law country

2.1.3 - Business

On the macroeconomic front, it carries news relating to GDP growth rates, unemployment and various government policies that affect the business environment. There is content related to new and upcoming startups that have entered the markets, various challenges, trends and success stories about them.

[illegible]

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The UK news segment starts with a top-down analysis of events happening within the UK. This could include major incidents, public events or any societal trends. The section opens with national news about the UK and then shifts to local news which is specific to the area of distribution for the newspaper. For this model, London newspapers have been picked up. London, being the financial hub of UK, captures significant news that could provide appropriate signals for the model.

Lastly it covers significant court cases, legal proceedings or law enforcement activities happening in the country. The section closes with the weather updates and environmental issues.



This section, covering the political news, talks about new events, policies and changes within the parliament and their impact on the public. There are articles which cover news about the major political parties - Conservative, Labour and Liberal Democrats - along with their leadership, ideals and agendas.

Raw word cloud for "Politics" articles

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news were included to facilitate the sentiment analysis in the right direction. The decision to exclude recreational news such as fashion, food, drinks and similar lifestyle sections was consciously taken to reduce the overall dimensionality of the problem thereby significantly reducing overall run-times for various algorithms.

2.2 - FTSE 100

Yahoo finance is one of the most comprehensive data source platforms which provides a wide range of financial information, including stock quotes, financial reports, business news, and investment resources. It is widely used by investors and finance professionals to understand stock market trends and make necessary investment decisions. Due to its wide scope, global acceptability and ease of access, Yahoo finance was considered to be a trusted data source and to provide accurate information for the purpose of training the model. Here, the “yfinance” API was used to call the FTSE 100 data and stored in a .csv file, to be used as a dependent variable during multilinear regression.

The FTSE 100 is the most recognized stock market index for the UK. It launched in 1984 consisting of 100 of the largest companies by market capitalization listed on the London Stock Exchange. These companies collectively comprise over 81% of the total market capitalization of the London Stock Exchange with a daily average trading volume of £5Billion. Moreover, 70% of the revenue generated by these companies comes from outside the UK thus the index, to an extent is also an indicator of international markets.

These 100 firms belong to various sectors: ⁵⁶

Sector	Percentage of FTSE 100 Index	Notable Companies	Market Capitalization (approx.)
Financial Services	19.07%	HSBC, Barclays, Lloyds Banking Group	£380-400 billion
Consumer Staples	17.09%	Unilever, British American Tobacco	£300-350 billion
Energy	12.82%	BP, Shell	£200-250 billion
Health Care	12.19%	GlaxoSmithKline, AstraZeneca	£150-180 billion
Industrials	13.18%	Rolls-Royce, BAE Systems	£100-120 billion
Materials	9.42%	Rio Tinto, Anglo American	£100-150 billion
Utilities	4.18%	National Grid, SSE	£50-70 billion
Consumer Discretionary	7.17%	Tesco, Next	£80-100 billion
Communications	2.79%	Vodafone, BT Group	£80-90 billion
Real Estate	1.02%	Land Securities, British Land	£30-40 billion

⁵ [FTSE 100 Index Sector Weightings | Sibilis Research](#)

⁶ [Sectors in the London Stock Exchange \(LSE\) | shareprices.com](#)

Thus, strong presence and wide industrial scope of the FTSE 100 companies make the index an apt choice as the economic barometer for the UK markets.

2.3. - The Dictionary

2.3.1 - Formation of the dictionary

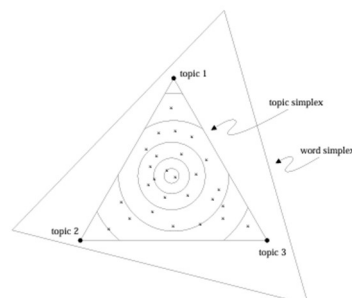
The dictionary played a key role in the model. It is a self-made dictionary, inspired by the NRC Lexicon and articles from 'Guardian', which were later used for training the model. The basic idea was to categorize the terms into 8 categories, where-in each category would represent a separate emotion. These categories were based on the research of [Plutchik \(1980\)](#) as these were the 8 core emotions which were expected to form a baseline for all other emotions, and thus would help in capturing all the necessary signals with respect to human emotions in the financial literature.

In the first stage, terms for each emotion were selected through manual inspection from the NRC Lexicon, such that which when paired with financial terms would hold meaning and provide signals with respect to the stock markets.

Once the base terminology was ready, Latent Dirichlet Allocation (LDA) was used to discover the most common financial terms that are used by journalists in various financial articles. LDA is a probabilistic model which was developed by [David M. Blei, Andrew Y. Ng and Michael I. Jordan \(2003\)](#), and published in the Journal of Machine Learning Research.

It is said to be a generative probabilistic model, since it assumes the distribution of the documents over various topics, whereby each topic is threaded to a bag of words. This provides an explicit representation of the document, and by passing the Guardian articles through this model allowed for topic discovery which aided in identification of the key terminologies that were used repeatedly in various articles and could provide signals for analysis.

Through their paper, David, Andrew and Michael worked out a variation of the pre-existing Probabilistic Latent Semantic Indexing (PLSI) model that was developed by [Thomas Hoffman \(1999\)](#). They came up with their own Dirichlet distributions, which helped in curbing the linearly increasing parameters in the case of the PLSI model. The PLSI was based on the assumption that occurrence of words in documents can be explained by latent (hidden) variables representing topics which further continued for LDA. However, unlike PLSI which results in a distribution of multiple topics over a single document, the Dirichlet distribution has a smooth function which assigns a singular topic for each document. This feature helps in discovering the otherwise hidden topics, when new and un-seen documents are passed through the model.



Journal of Machine Learning Research 3 (2003) 993-1022

There are eight categories that were used, namely: joy, trust, fear, surprise, sadness, anticipation, anger and disgust. Each of these was expected to have a degree of correlation with stock market reactions which show up in news articles, before the stock market reacts.

A joyous emotion in the news articles tends to indicate a positive sentiment within the stock markets which results in increased volumes and heavy buying on the exchange. This is usually triggered through bullish news about companies such as earnings release, favorable economic policies and other such news indicating an optimistic future. Articles expressing joy often use words like "growth", "record profits", "expansion", "booming," and "success."



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Surprise is a direct opposite of the emotion “trust” and is associated with high volatility and unstable stock markets. It could include articles pointing in both positive or negative direction, such as unexpected profits during an earnings release, an unexpected change in interest rates by the Bank of England or surprise mergers and acquisitions. Some of the commonly used words in such articles are “unexpected”, “shock”, “unforeseen”, “sudden” and “surprise”.



Sadness correlates with a negative sentiment within the stock markets thus leading to decline in the price levels for the company or the economy in question. Prolonged periods of such a negative sentiment may lead to investors diverting their funds to different sectors or asset classes. This can be seen through articles covering companies reporting losses, layoffs or long-term industry decline. It can be identified through some common words such as "loss", "decline", "struggles", "disappointing," and "downfall".



Anticipation often leads to speculative buying within the markets. Investors try to position themselves based on their expectations of upcoming news. The direction of move often depends on the wholistic sentiment of the market before such releases. This behaviour is seen before upcoming earnings release, product launches, release of economic data or any decisions about to be made on upcoming policies. Post release, in case of an unexpected result (surprise), there is significant volume in the opposite direction. Anticipation can be identified through words such as: "expected", "forecast", "projected", "awaited" and "speculation".



Anger is deemed to be an indicator of negative sentiment in the markets and can often cause public backlash & regulatory scrutiny. This can be seen through articles carrying information about Companies embroiled in scandals, unethical practices, corporate misconduct, executive misbehaviour, or controversial business practices. Articles depicting anger carry terms like "outrage", "scandal", "backlash", "fury" and "protest".



Articles characterized by “Disgust” indicate a negative sentiment within the markets. This can lead to severe reputational damage for the company and thus investors tend to distance themselves from a particular sector or industry. It can be seen through articles focussing on major scandals, environmental destruction or extreme corporate malpractice. Companies may also experience prolonged boycotts or price dips following incidents that may provoke widespread disgust as natural disasters or serious ethical violations. Articles expressing disgust tend to carry words like "repulsive", "disgrace", "abhorrent", "unacceptable" and "shameful".



Section 3: Methodology

This section discusses the text analytics approach that has been used in this study, coupled with an unsupervised method followed by regressions. The model was constructed in two phases.

The first phase involves preprocessing text and calculating similarity scores between the articles from the 'Guardian' obtained via the API and the dictionaries made for each of the eight emotions mentioned in the foregoing section. This resulted in eight different similarity scores for each trading day. Here it was necessary to work on a trading day basis to keep the data structure same across all variables. The similarity scores were calculated using four different methods to ensure that all the necessary signals are completely captured. The methods being: Jaccard, Levenshtien, Cosine theta and Euclidean distances.

The second phase of the model utilized Principal Component Analysis which takes the four similarity scores for each emotion as input into a separate model, thus resulting in eight outputs, each being the first principal component for every emotion.

These Principal Components were then passed through a lagged multilinear regression model whereby the best lag was selected based on a loss function, and the loss function was picked on the basis of predictability accuracy. This accuracy was based on a simple statistic, i.e. maximization of the number of times the predicted and actual returns lie on the correct side (positive or negative).

3.1 - Pre-Processing

Preprocessing is the first and most important step when using text analytics. Being a very high-volume dataset, consisting of over eight hundred thousand news articles, it becomes difficult to process the entire dataset as it is. Thus, various methods and techniques were employed to eliminate the noise from the data and retain only those key signals which may be necessary for the model to capture the relevant information.

The first step in preprocessing was to filter the text and remove the Stop-Words. Stop-Words refer to the sect of words in the English vocabulary which hold no meaning in themselves, however, are necessary for accurate grammatical representation of text in an article. These carry no relevant signals with respect to the dependent variable in question and must be removed when performing NLP (Natural Language Processing) or Text analysis, as it is necessary to eliminate this noise to capture the signal more accurately. This task was done using a pre-existing package within the 'nltk' library of python. The 'nltk' library in python helps remove this noise by filtering out articles, prepositions, pronouns, conjunctions, common verbs such as: 'is', 'are', 'was', 'were' etc. and other such common words like: 'of', 'with', 'as', 'for' etc..

Removal of these words helped in significantly reducing the dimension of the data. This further aided in reducing overall runtimes of the similarity algorithms and improved accuracy by allowing the model to appropriately capture the signals more effectively.

Given that the model uses over 800,000 articles as input, that still leaves a significant amount of textual data with unwanted noise. Thus, to further filter for the key signals, part of speech tagging was employed. Since the goal was to compare the article with a dictionary, only those POS (Parts of Speech) tags were retained in the text, which were present in the dictionary designed.

Part of Speech tagging refers to identifying the form of the term within the English language, such as a verb, noun, adjective, article, supporting verb and so on. For this model, verbs, adjectives and

nouns have been included as these forms of speech contain important information from the perspective of understanding the presence of an emotion.

The dictionary in use consists of terms with single words to phrases of up to three words. Thus, it was essential to ensure that the processed data also follows a similar structure for the similarity algorithms to work accurately. For this purpose, 'n-gram' approach was deployed with 'n' running from one up to three. Additionally, from a theoretical standpoint, 'n-gram' approach with terms from one up to three words holds more weight, since the structure of the signal words is expected to be either a stand-alone word depicting a particular emotion, or the emotional term combined with financial terminology which may be in the form of a phrase of up to three words.

The final step in pre-processing was stemming the terms to their root form. This was done using the porter stemmer library readily available within python. The stemming algorithm was developed by Martin Porter (1980). The objective of the algorithm is to strip the word of any prefix and suffix attached to it to bring it to the root form. This is done in five steps.

It starts by removing any plural, past or continuous tense suffix present in the word. In case of plural words, it eliminates the 's' or 'es' in the end, such as 'ponies' would become 'poni'. In case of past or continuous, it eliminates the 'ing' or 'ed' present at the end of the word such as: 'running' would become 'run' and 'played' would become 'play'. In case of any words ending with a 'y', the letter gets swapped for an 'i'. this was added to handle words such as 'happily'. Thus 'happy' would become 'happi', 'cry' would become 'cri' and so on.

In the next three steps, the algorithm further focuses on removing suffixes over three phases to reach the core form of the word. The first would be to eliminate suffixes like 'ational', 'ization', 'fulness'. In the second step it would remove suffixes like 'ative', 'alize' etc. and the third would remove any remaining suffixes like 'ement', 'ment' etc. a word like 'formalization' would thus become 'formal' after going through the first two phases. The final phase works on words such as 'dependency' which would become 'depend' to remove any remaining suffix. In its final step it removes the 'e' in the end if there is any when the word is longer than a certain length or follows a particular sequence.

Here, stemming was chosen over lemmatization to arrive at the core form of the word despite accuracy, due to the volume of the data. Stemming is a more aggressive method and much faster when dealing with larger datasets. Additionally, research has shown marginal differences between the two when looking at the final results.

3.2- Similarity Algorithms

The main weight of the model falls on the four key similarity algorithms used to analyze the text and capture presence of specific emotions that may drive investors to make investments accordingly. The structure of the model is designed in such a way that it uses a stacking technique on the results of all four similarity algorithms being performed. The reason for not being restricted to one method is the difference in structure of each of these methods. Due to the difference in how the algorithms calculate and quantify similarities with respect to dictionaries allow each one of them to capture a different form of signal present in the text.

3.2.1 - Levenshtien Distances

Levenshtien distances was a method developed in 1966 by Vladimir I. Levenshtein (1966) which calculates similarities between two document strings based on the minimum number of single character edits needed to bring one vector on the same plane as the other. The algorithm was initially designed to fix errors that arose during data transmission due to noisy lines and was the first of its kind that could handle not only substitutions, but also insertions and deletions to correct the data.

The Levenshtien algorithm works by calculating a global minimum of the changes required to convert one string into another through insertions, deletions and substitutions by carrying out all possible sequences and combinations to reach the said minimum. For every edit needed, it adds a distance of one between the two strings.

The edit distance can be converted into a ratio which works between the range of '0' to '1', where 1 would mean perfect similarity and 0 would indicate no similarity between the two vectors. Since Levenshtien distances work inversely to similarity scores, for the purpose of this model, Levenshtien ratios have been used, which is a mere inverse of the edit distance. Hence, the higher the ratio, higher is the similarity between the two vectors. The model in use calculates the Levenshtien ratio using the readily available python library for all combinations of unigrams, bigrams and trigrams with every term within the dictionary and then returns an average ratio for all the combinations for a given day.

3.2.2 - Cosine Similarity

Cosine theta similarity was introduced by Gerard Salton, Michael J. McGill (1983) and within their text, they focus heavily on vector space models using Cosine similarity as a crucial method to measure document similarity. Cosine theta similarity is further introduced in the literature to be used as a measure of similarity between two latent semantic vectors to perform Latent Semantic Analysis since unlike the vanilla Cosine measure, Cosine theta ignores the length of the document and only focusses on the angular similarities.

These scores based on the angles between the two vectors are calculated through the Cosine of the angles between the two different terms to check for the level of similarity between them. The cosine similarity between two vectors A and B is computed as the dot product of the vectors divided by the product of their magnitudes:

$$\text{cosine similarity } (A,B) = (A \cdot B) / (\|A\| \|B\|)$$

This results in a similarity measure that ranges from -1 to 1, where:

- 1 indicates that the two vectors are identical in orientation.
- 0 indicates orthogonality or no similarity.
- -1 indicates completely opposite directions.

The model aims to calculate similarity between a pre-established dictionary and every individual article, where the length of the dictionary is significantly greater than that of any individual article thus to overcome this, Cosine Theta was preferred as a measure of similarity. This was done by first combining the dictionary for each emotion into a single string, thus resulting in 8 different strings. Cosine Theta of the angle between these string vectors and n-grams for each date were then calculated separately to arrive at the similarity scores for each date.

3.2.3 - Jaccard Similarity

Jaccard is a 20th century measurement for similarity, that was developed by Paul Jaccard (1901) who was a Swiss botanist researching the diversity and distribution of plant species within the alpine zone. He observed that plants from different regions had varying levels of similarities amongst them and thus to compare these similarities across regions, he developed a coefficient to compare the composition of these species. It works on the mechanics of set theory evaluating the similarity between finite sets by dividing the size of the (shared species) intersection by the size of the union of the sample sets (total number of species in the region):

$$J(A, B) = (|A \cap B|) / (|A \cup B|)$$

Where $|A \cap B|$ is the number of elements common to both sets,

and $|A \cup B|$ is the total number of elements in either set.

A simple example of its working can be seen in the following:

Consider two text documents:

- Document 1: "the cat is on the mat"
- Document 2: "the mat is under the cat"

The unique words in each document are:

- Document 1: {the, cat, is, on, mat}
- Document 2: {the, mat, is, under, cat}

The intersection of the two sets (common words) is {'the', 'cat', 'is', 'mat'}, and the union (all words) is {'the', 'cat', 'is', 'on', 'mat', 'under'}. The Jaccard Similarity thus is: $J = 4/6 \approx 0.67$.

The model initializes calculating Jaccard similarity by first creating individual strings of documents for each date and separate strings for each of the emotions being analysed. The document string is then converted into a matrix using the term frequency – inverse document frequency calculation. This is preferred over a simple frequency matrix as this adds weight to words that are present uniquely in a particular document and penalizes for terms that occur very often across the corpus. This technique helps in isolating the signal words. Post this pre-processing, once the data is ready, the 'sklearn' library is used to compute Jaccard similarity scores, which is an intersection between the dictionary and the documents divided by the total pool, done separately for each day.

3.2.4 - Euclidean Distances

Euclidean distance is a concept derived from Euclidean geometry which measures the "straight line" distance between two points in a multidimensional space, developed by Euclid (300 BCE). This concept introduced by Euclid was further enhanced by Salton, Gerard, Wong, A., & Yang, C. S. (1975) where they extended the standard geometric Euclidean distance to calculate document similarities through vector space models. In a vector space mode, each dimension of the vector corresponds to a

specific term in the vocabulary, and the value in that dimension represents the term's frequency or any such numerical statistic. The most commonly used is the term frequency – inverse document frequency (tf-idf) which was introduced by Salton, Gerard, Wong and Yang in their paper. After representing the documents and queries as vectors in a high dimensional space, they used Euclidean distances to measure how close the documents are to queries.

In the context of text analytics, it helps quantify the dissimilarity between documents, representing each term as a point in the high-dimensional space.

For this model also, a tf-idf approach has been employed. The dictionaries for each of the emotions are combined into a single string and similarly, documents for each day are combined to a string. The dictionaries and documents are both then represented in a high dimensional space through term frequency – inverse document frequency vectorization to compute the Euclidean distance for each emotion per trading day. However, in the end, once distances are computed, these are then inversed to allow for proportionality with similarity between the documents and dictionaries.

3.3 - Predictive Model

Once the similarity scores are calculated through each of the above similarity algorithms, they are passed through a Principal Component Analysis (PCA) model which returns the first Principal Component for each emotion. These Principal Components then go through a lagged multilinear regression model which helps in predicting the index returns.

3.3.1 Principal Component Analysis

Principal Component Analysis was first introduced by Karl Pearson (1901). Pearson's research was directed towards understanding complex data in a relatively simpler way, i.e. he was looking for a way through which multi-dimensional data could be expressed using a smaller subspace, while maintaining significant explainability at the same time. The paper follows a geometric approach. He uses a set of data points on a multidimensional space and tries to represent the same using a smaller subspace, while minimizing the sum of least squares, which forms the idea of "closest fit" for all regressions.

He discovered that the first principal component corresponds to the direction in which the data varies the most, and it is the line that minimizes the sum of squared perpendicular distances from the points to the line. This idea formed the base for Harold Hotelling (1933) who went on to introduce the concept of eigenvalues and eigenvectors in the context of PCA. Hotelling's work was driven by the need for reducing data dimensionality due to the significant advances in data collection techniques around the 20th century, and complexities which arose in managing this data. In his paper, he shows that Eigenvectors of a multidimensional space determine the direction along which the data varies, and Eigenvalues defined the magnitude of this variability for the corresponding Eigenvectors. Thus, the principal component with the highest Eigenvalue explained the maximum amount of variability in the data.

This concept has been adopted here with the objective of extracting the key Principal Component for each emotion. Since each method calculates similarities differently, the form of signal carried by each of the scores is different, creating a need to combine these signals.

Principal Component Analysis was performed separately on the similarity scores obtained for each of the eight emotion scores through Jaccard Similarity, Levenshtien Distances, Cosine Similarity and Euclidean Distances and the first Principal Component was selected from each of the eight PCAs.

The results from each of the eight PCAs were further used as an input for a lagged multilinear regression, where the lag was selected based on lowest loss score to provide the best fit to the model. The dependent variable here is the FTSE 100.

3.3.2 Multilinear Regression

The concept of Multilinear Regression is an extension of the Linear Regressions that was first introduced by Francis Galton (1886) while he was analyzing the relationship between the heights of parents and their offsprings. He discovered that children of particularly tall or short parents tend to have heights closer to the average population, which lead to the concept of “regression towards mediocrity”.

This was later formalized by Karl Pearson (1908), where he expanded Galton’s idea of linear regression to the dependent variable having relationships with multiple input variables separately. He and also introduced the concept of multiple correlation coefficient, which explains how well the individual variables put together are able to explain the dependent variable.

The above idea introduced by Karl Pearson has also been applied within this model, to establish the relationship between the principal components extracted for each of the emotions through PCA analysis and the FTSE 100 index, which has been used as a benchmark for the UK stock market price levels.

Additionally, a lag has been added to the model in the input variables, since it is believed that the input variables show signals before the output variable takes upon it’s expected value. This lag is selected based on a loss function, and the loss function to be used is determined based on best predictability for the model. The predictability of the model is determined on the basis of an agreement rate statistic, which calculates the proportion of time the predicted values and true values lie on the correct side, i.e. both indicate positive returns or negative returns.

The three loss functions that were used were:

- a. Mean Absolute Error (MAE)
- b. Mean Squared Error (MSE)
- c. Huber Loss

The Mean Absolute error looks at the average absolute difference between the predicted and true value. This method is more direct and gives the same weight to all deviations while selecting the best lag.

Mean Squared error uses the average squared differences between the two values and selects the best lag accordingly for each variable, to incorporate within the model. The MSE is beneficial when more weight needs to be given to extreme value errors and optimize the model accordingly.

The third method used is the Huber Loss function, which comes from Peter J. Huber (1964). Peter Huber found a way of estimating the central moment in question through a method which is less sensitive to outliers or abnormalities. But, at the same time, still takes them into account while

modelling. The key idea behind his approach was a compromise between L2 (least squares) and L1 (absolute deviation) norms.

In the end, the Means Squared Error took precedence over the others, since the main objective was to model and predict outliers on the left tail of the distribution. The method's property of squaring the distances gives more weights to extreme values and improves the effectiveness of the model to predict these extreme negative returns.

3.4 - Model Validation

3.4.1 - Philips, Shuping and Jun Yu (PSY) Test

The Philips, Shuping and Jun Yu Bubble Test was developed to check for mild explosiveness within an Auto Regressive (AR) model, by slightly modifying the Augmented Dickey Fuller test so that it works recursively, i.e. for a given end point in time, the algorithm would appropriately pick an appropriate starting point.

The test works under the null that there exists martingality within the AR model, however with a drift component. Algebraically it can be shown as:

$$Y_t = gT + y_{t-1} + \mu t$$

Where the term "gT" is used to add stochasticity to the model and indicate a deterministic trend.

The reason behind using a recursive algorithm instead of a rolling or expanding window was due to its efficiency and effectiveness which has been discussed in the paper "Real Time Monitoring of Asset Markets: Bubbles and Crisi

s" by Peter C.B. Phillips, Shuping Shi, and Jun Yu (2015). The algorithm works by fixing an end point assuming r_{end} and let r_{min} be the minimum window to be able to run the regressions. The recursive algorithm picks r_1 such that it lies between 0 and " $r_{\text{end}} - r_{\text{min}}$ ". Thus, the ADF is calculated as:

$$P_{SYr} + (r_{\text{min}}) = \sup \{1 \in [0, r_{\text{end}} - r_{\text{min}}]\}, r_{\text{end}} = r + ADFr_{\text{end}} r_1$$

The supremum function here, allows to select the most efficient start point for the recursive algorithm.

Once the recursive window is generated and the ADF t-statistic is calculated (based on the critical value picked, which in this instance is 5%), regions wherever the null hypothesis gets rejected would indicate mild explosivity in the prices and indicate formation of bubbles.

This method was run parallelly to cross check and help verify if such explosiveness can be modelled through the textual approach mentioned in this study.

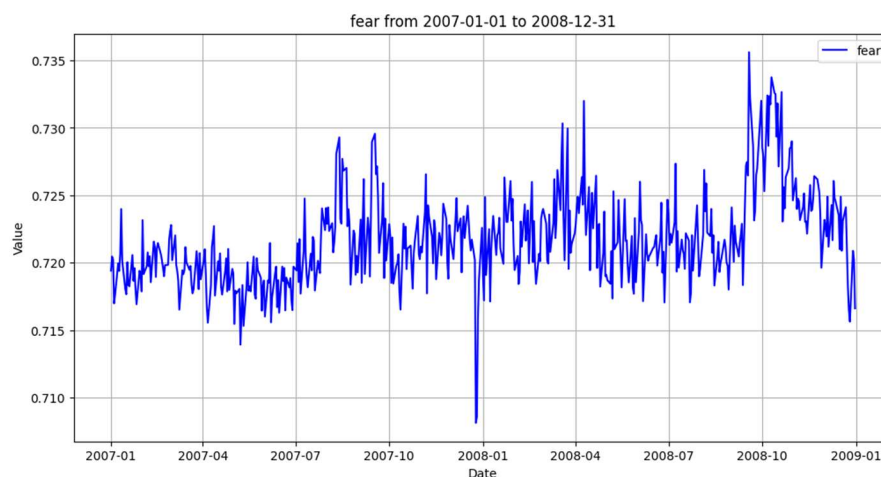
Section 4:
Discussion of Results

This section discusses the output generated through the financial model built in the python environment. Over the past 20 years, UK has experienced several stock market crashes that have been driven by global financial events. The model aims to predict these in advance and verify using the PSY bubble test.

Overall, the results show a strong correlation between the emotions extracted from the Guardian news articles and returns on FTSE100. To validate the workings of the model, four key market crashes have been used: 2007-08 housing bubble crisis, 2011-12 sovereign debt crisis, 2015-16 Brexit Referendum and 2020 Covid. During each of these events, articles have shown strong variations in emotions which have been analyzed below through their principal components.⁷

4.1 - Housing Bubble Crisis (2007-08)

This was one of the most severe financial crisis which originated from the housing bubble in the United States real estate markets. It had multiple layers of derivatives on the sub-prime mortgages over and above their true value, which magnified the returns generated by the underlying asset. Hence, when the bubble burst, it set off a chain reaction de-valuing all the derivatives, leading to significant financial losses globally.⁸



Analyzing fear, as seen in the first principal component for fear plotted, the level shifted upwards from 2007/07, as this was the period when sub-prime mortgages exited the first phase of low instalments, and the number of defaults gradually started climbing due to the increased monthly payments attached to the second phase.

In early 2008, there were two major factors that dropped the level of fear seen within the stock markets. Firstly a 75-basis point rate cut made by the US Federal Bank, which was the largest ever cut made in history⁹. The Federal Bank also announced its readiness to take further measures, when necessary, to ensure economic stability. Secondly, President George Bush announced an economic stimulus package which though did not directly target the sub-prime crisis but included tax rebates

⁷ https://en.wikipedia.org/wiki/List_of_stock_market_crashes_and_bear_markets

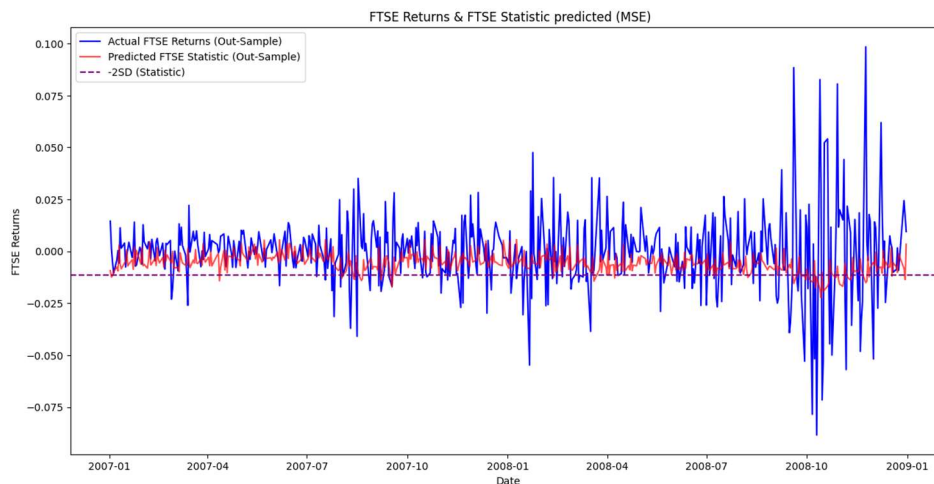
⁸ <https://www.earn2trade.com/blog/michael-burphy/#:~:text=When%20Burry%20bought%20his%20first%20CDS%20on%20mortgage,for%20being%20the%20first%20to%20ever%20buy%20them.>

⁹ <https://www.federalreserve.gov/newsevents/pressreleases/monetary20080122b.htm>

for individuals and incentives for businesses to invest and hire. This was done to boost consumer spending and confidence within the economy.¹⁰

However, despite these measures taken by the US Government, the level of fear in the equity markets increased and hit a high towards the end of September 2008. This is when the faults within the structure of credit rating agencies was revealed.

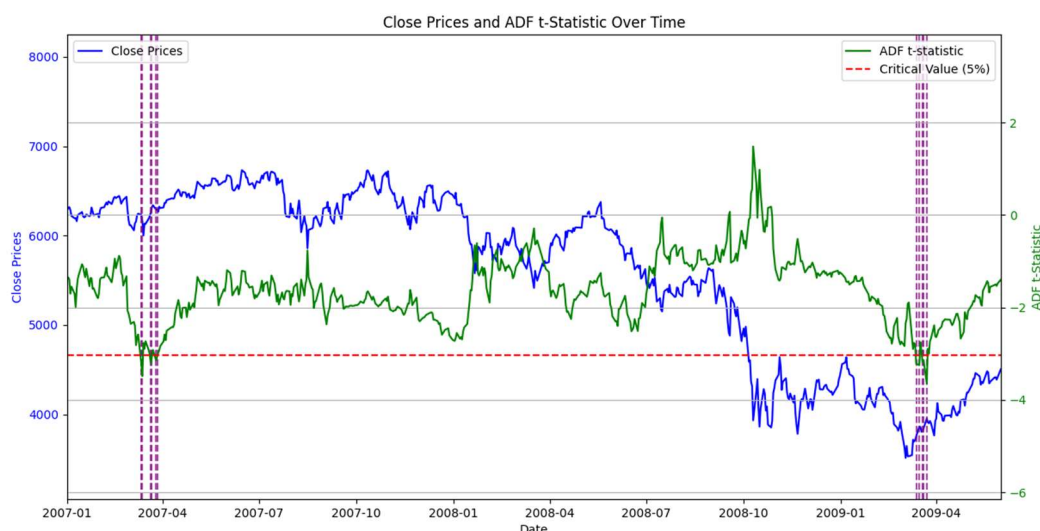
The level dropped close to its mean value in 2009, once the economy and stock market indices started to recover.



Looking at the statistic generated using the Principal Components for each of the emotions, abnormally negative returns were seen when the predicted statistic crosses below its two standard deviation mark. Below is a list of date ranges whereby the statistic exits the two standard deviation mark, and the market experiences negative returns.

Start Date	08/02/2007	08/13/2007	08/20/2007	09/07/2007	09/17/2007	07/08/2008	08/08/2008
End Date	08/02/2007	08/14/2007	08/21/2007	09/07/2007	09/17/2007	07/08/2008	08/08/2008
Start Date	09/19/2008	09/26/2008	10/02/2008	10/09/2008	10/16/2008	10/27/2008	10/30/2008
End Date	09/19/2008	09/30/2008	10/07/2008	10/14/2008	10/21/2008	10/28/2008	11/04/2008
Start Date	11/07/2008	11/24/2008	11/27/2008	12/18/2008	12/30/2008		
End Date	11/10/2008	11/25/2008	11/27/2008	12/18/2008	12/30/2008		

¹⁰ <https://georgewbush-whitehouse.archives.gov/news/releases/2008/01/20080118-1.html>

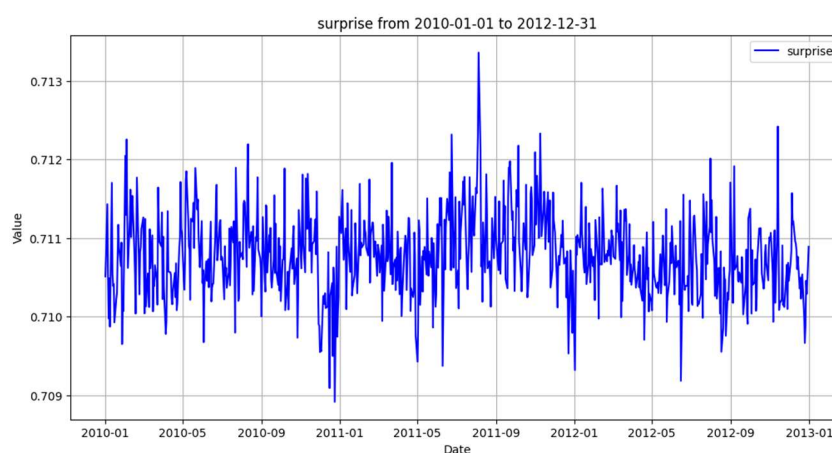


Observing the ADF statistic at a 5% significance level with a critical value of 3.04 in the graph above, there was mild explosiveness in the model observed during March 2007 and March 2009. Though these signals of explosiveness were very delayed, they do verify non-stationarity within the model and presence of bubbles in the prices. The recursive Augmented Dickey Fuller test identified the date ranges given below as having mild explosiveness in the Auto Regressive (AR) model for prices.

Start Date	03/13/2007	03/21/2007	03/22/2007	03/26/2007	03/27/2007	03/13/2009	03/19/2009	03/20/2009
End Date	03/13/2007	03/21/2007	03/22/2007	03/26/2007	03/27/2007	03/16/2009	03/19/2009	03/23/2009

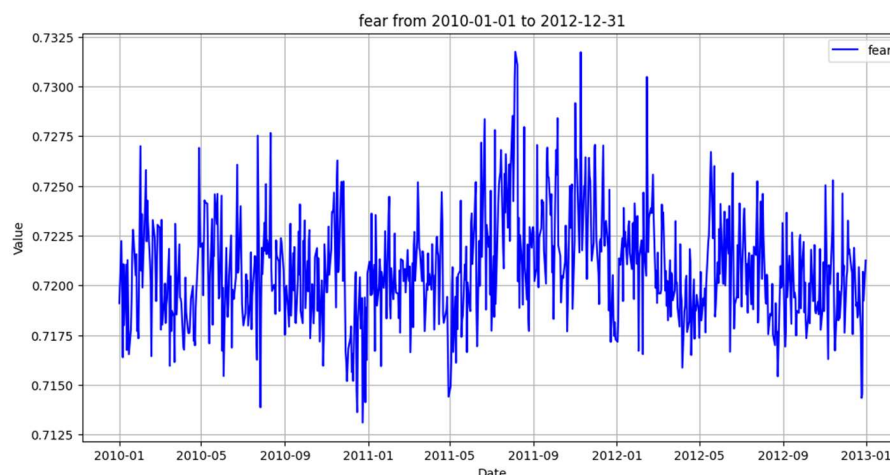
4.2 Eurozone Sovereign Debt Crisis (2011-12)

The Eurozone debt crisis was said to be an aftershock of the 2008 bubble, which started when Greece significantly understated its budget deficit for years and its borrowing costs skyrocketed, making it impossible for the country to refinance its debt. During this period, UK banks had a total exposure of approximately £100 billion as reported by the Bank of International Settlements statistics. Even though the direct exposure to Greek sovereign bonds was limited to £8 billion, there was significant exposure in Ireland, summing up to £85 billion due to close economic ties. The Irish economy felt the shock close to summer 2011 with soaring bond yields that threatened financial stability.¹¹



¹¹ <https://data.bis.org/topics/IDS/data>

Analysing “surprise” prevalent in the market during this period, a sudden spike in the emotion was observed during August 2011. On August 5, 2011, Standard & Poor’s downgraded United States’ long-term credit rating from AAA to AA+ for the first time in history due to the US Government’s political gridlock over raising the debt ceiling and addressing national debt. This led to significant volatility in the global markets, and since UK banks held significant exposure to US equities and derivatives, this came as a surprise for the economy amidst the Eurozone debt crisis.



The level of fear within the UK economy started shifting upwards after May 2011 when the shocks of the Greek debt-crisis reached Ireland, where UK banks had significant exposure.

In August 2011, there was a fall in the level of fear within the economy, when Eurozone declared a bailout package amounting to €109 billion for Greece. This included measures such as lowering interest rates on existing loans and extending the repayment periods. Though this was a crucial step in aiding Greece to repay debts, it had to be combined with stringent economic measures within the country for it to come out of the crisis, which lead to social unrest within the country thus spreading fear close to November 2011.

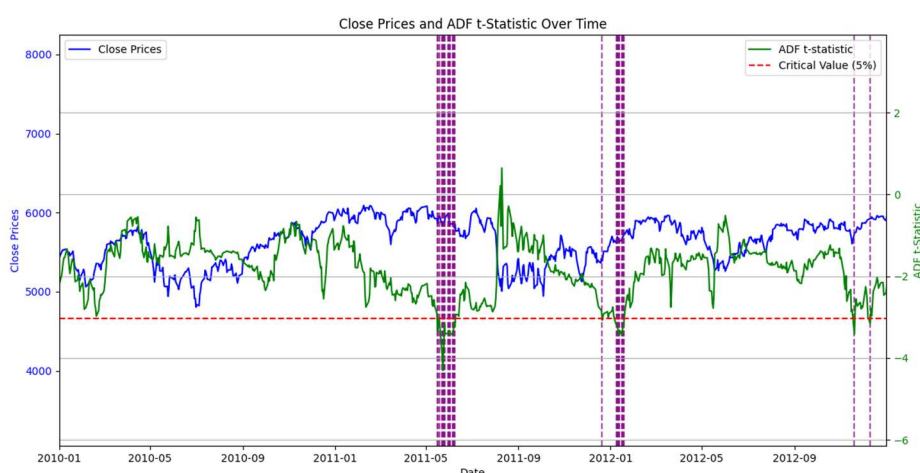
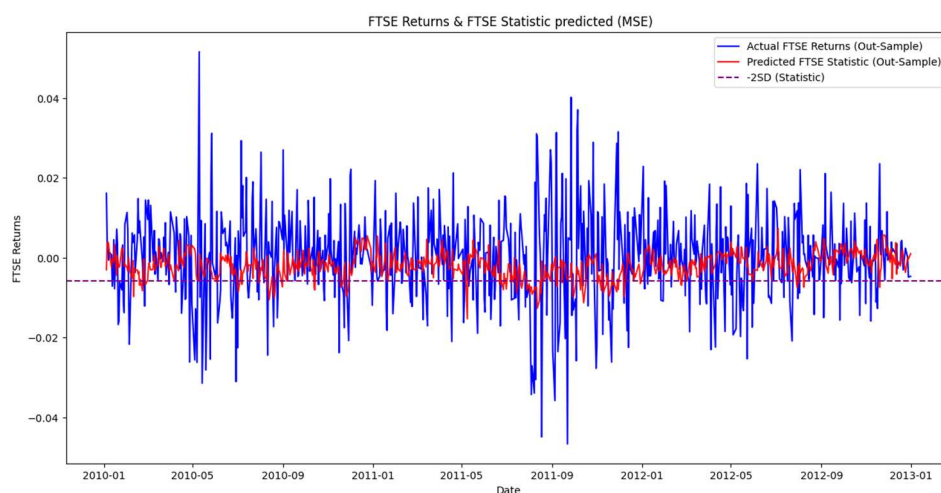
In December 2011, the EU agreed to adopt stricter fiscal rules, known as the Fiscal Compact. This treaty focussed on countries maintaining balanced budgets and introduced sanctions for countries that were found violating these rules. This helped restore temporary confidence within the European economy.

Lastly, in March 2012, the European Stability Mechanism was established with a lending capacity of €500 billion and was structured to provide support to crisis facing nations. This marked an end to the debt-crisis and can also be seen in financial literature through the level of fear being restored to their mean values.

The expected statistic generated using Principal Components for each of the input emotions showed results in tandem with the returns. Significant negative returns can be seen during the periods where the statistic breaches the two standard deviation mark. The date ranges for predicted negative returns are as below:

Start Date	05/18/2011	06/06/2011	06/30/2011	07/05/2011	07/07/2011	07/14/2011	07/19/2011
End Date	05/18/2011	06/06/2011	06/30/2011	07/05/2011	07/07/2011	07/14/2011	07/19/2011
Start Date	08/09/2011	08/24/2011	08/26/2011	08/30/2011	09/19/2011	10/04/2011	10/12/2011
End Date	08/12/2011	08/24/2011	08/26/2011	09/08/2011	09/19/2011	10/04/2011	10/12/2011

Start Date	10/20/2011	10/25/2011	11/01/2011	11/03/2011	11/24/2011	11/29/2011	12/02/2011
End Date	10/21/2011	10/25/2011	11/01/2011	11/04/2011	11/24/2011	11/30/2011	12/02/2011
Start Date	12/07/2011	12/13/2011	12/15/2011	12/20/2011	12/29/2011	01/03/2012	
End Date	12/08/2011	12/13/2011	12/15/2011	12/20/2011	12/30/2011	01/04/2012	



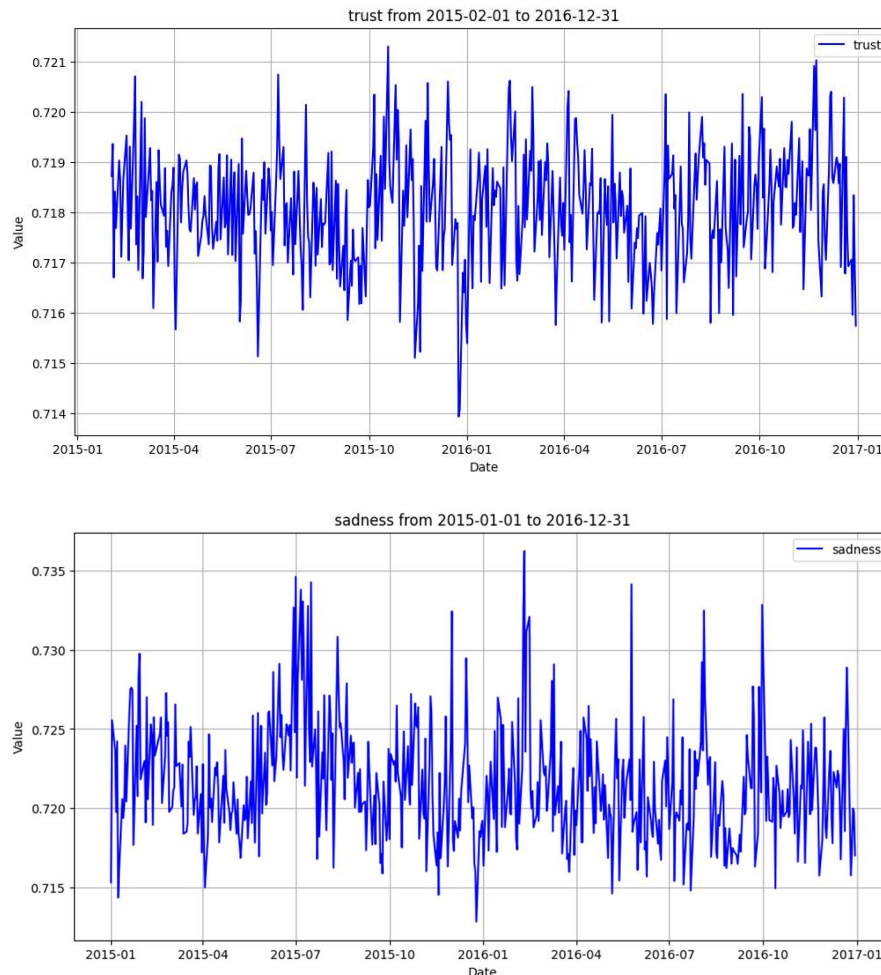
The above graph shows the recursive Augmented Dickey Fuller t-statistic in green and plots vertical lines to mark points in time where the statistic falls outside of the two standard deviation mark. Philips Shipping Shi and Jun Yu's ADF model is seen to be successful in identifying the upcoming crisis and helps in cross verifying the accuracy of the predicted statistic.

Below are the date ranges where the Autoregressive model for prices was seen to move away from stationarity and showed signs of mild explosiveness.

Start Date	05/17/2011	05/18/2011	05/20/2011	05/24/2011	05/25/2011	05/26/2011	05/27/2011
End Date	05/17/2011	05/18/2011	05/23/2011	05/24/2011	05/25/2011	05/26/2011	05/31/2011
Start Date	06/01/2011	06/02/2011	06/03/2011	12/21/2011	01/09/2012		
End Date	06/01/2011	06/02/2011	06/09/2011	12/21/2011	01/19/2012		

4.3 Brexit Referendum (2015-16)

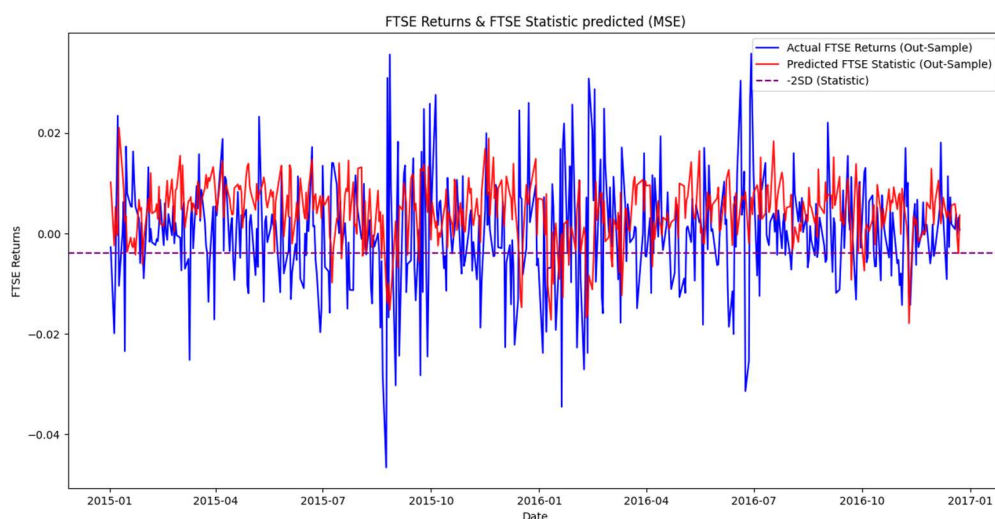
The 2015-16 Financial crisis which hit the UK markets significantly was an outcome of the Brexit referendum signed by David Cameron. This decision to exit the European Union led to significant economic uncertainty, social unrest and high stock market volatility within the country. This unrest could be seen through the different articles published in financial literature as discussed below.



The first signs of a negative sentiment (sadness) were visible in July 2015, after David Cameron came into power and decided to uphold the Conservative's manifesto of exiting the European Union as this introduced a degree of uncertainty within the markets leading to a cautious approach in investments. Towards the end of 2015, there were significant doubts about Brexit and its impact on the economy which can be seen through the financial literature indicating a significant fall in trust in the system as seen in the graph.

This was later followed by waves of negative sentiment within the financial markets as seen in the graph for "sadness". This could be seen in articles during February 2016 after David Cameron announced the date for the Brexit referendum. This announcement set stage for months of intense campaigning by both the "Remain" and "Leave" camps. On the final day of the referendum, there was a high magnitude of negative sentiment as the polling results were in favour of remain however, the referendum concluded with 52% in favour of leaving the European Union which shook the UK economy.

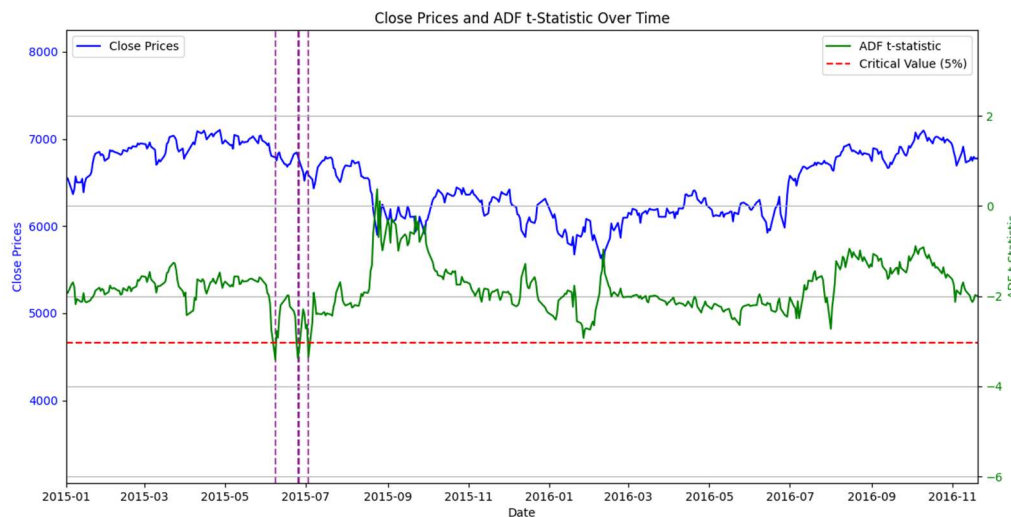
Furthermore, in 2016 October, Prime Minister Theresa May outlined her approach to Brexit and stated that the formal process for leaving the EU, would be triggered by the end of March 2017. This speech signalled a strong move towards Brexit and raised concerns about potential for new trade barriers with the EU causing increased market volatility. The Great Britain Pound experienced a significant decline of 6% against the United States Dollar during Asian market hours, leading to a strong negative sentiment, also visible in the financial articles (sadness).



Observing the graph above, the predicted statistic crossed the two standard deviation line soon after David Cameron came into power, which indicated that Britain was headed towards Brexit and again when the date for Brexit referendum was announced. This could be seen via the returns which followed through on the FTSE 100 exchange.

According to the predicted statistic, the following date ranges

Start Date	07/09/2015	07/17/2015	08/04/2015	08/20/2015	08/24/2015	09/02/2015	09/11/2015
End Date	07/09/2015	07/17/2015	08/04/2015	08/20/2015	08/28/2015	09/03/2015	09/11/2015
Start Date	09/21/2015	12/15/2015	01/08/2016	01/11/2016	01/13/2016	01/18/2016	01/21/2016
End Date	09/21/2015	12/17/2015	01/08/2016	01/11/2016	01/13/2016	01/18/2016	01/21/2016
Start Date	01/26/2016	02/05/2016	02/10/2016	02/15/2016	03/02/2016	03/09/2016	03/11/2016
End Date	01/26/2016	02/05/2016	02/12/2016	02/16/2016	03/03/2016	03/09/2016	03/11/2016
Start Date	04/06/2016	09/22/2016	10/03/2016	11/10/2016	12/22/2016		
End Date	04/06/2016	09/22/2016	10/03/2016	11/11/2016	12/22/2016		



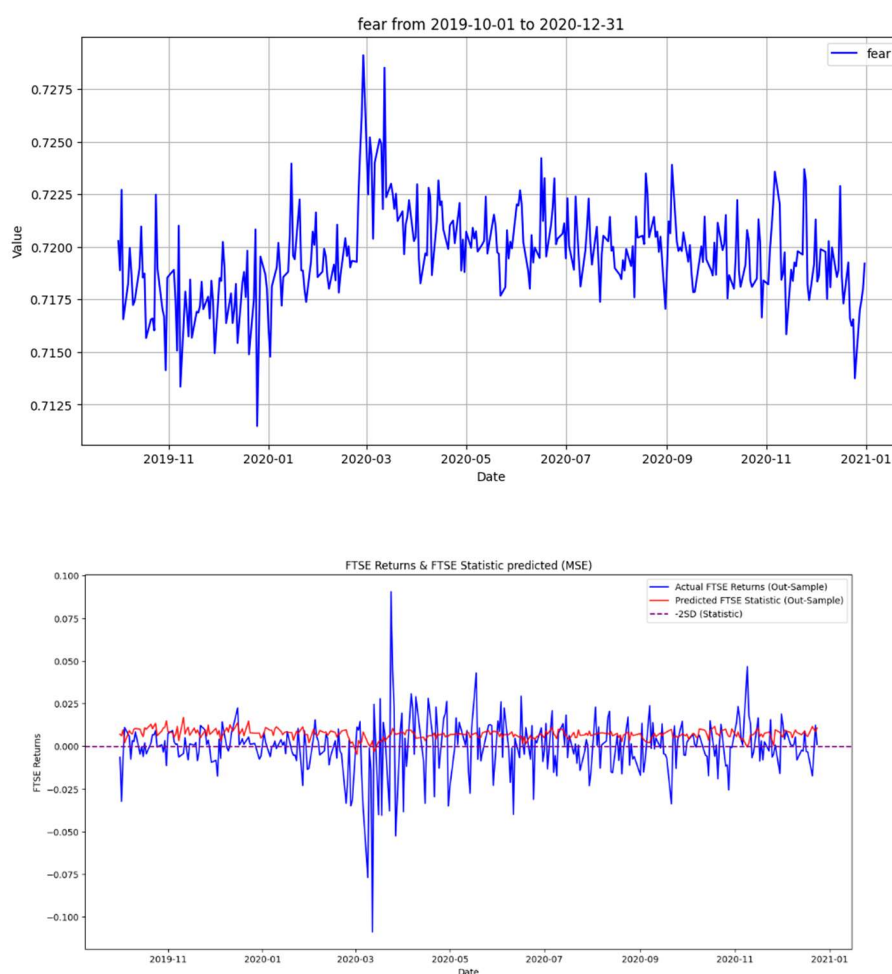
The above graph shows the PSY Bubble test, which uses a recursive ADF t-statistic based on a 5% tolerance level. The test has successfully identified explosiveness during the period of election results indicating a bubble, however, seems to have failed to predict the negative returns post Brexit. The t-statistic was close to the value of significance and would have been captured at a 10% significance, which has not been chosen here to avoid multiple false signals and because of the general acceptability of 5% in econometric literature. This explosiveness was identified in the following date ranges.

Start Date	06/08/2015	06/25/2015	06/26/2015	07/03/2015
End Date	06/08/2015	06/25/2015	06/26/2015	07/03/2015

4.4 COVID'19 (2019-20)

Unlike the above analysed financial crisis, 2020 experienced a widespread pandemic of the COVID virus which was discovered in Wuhan, China in December 2019 and said to have been spreading since October 2019. This virus spread globally forcing international governments to prohibit and/ or restrict international travel and to impose curfews and lockdowns within their own nations to prevent this virus from spreading. This led to a halt in business activities globally and thus the impact could be seen on the financial markets.

As observed in the similarity scores for fear below, the level of fear was relatively low towards the end of 2019 due to assumptions that the disease had been contained within China. However, the similarity scores started rising towards the beginning of 2020 and hit peak in March 2020, which is when the British government imposed its first nationwide lockdown. The first lockdown lasted for six weeks after which, the economy was gradually allowed to open in phases.



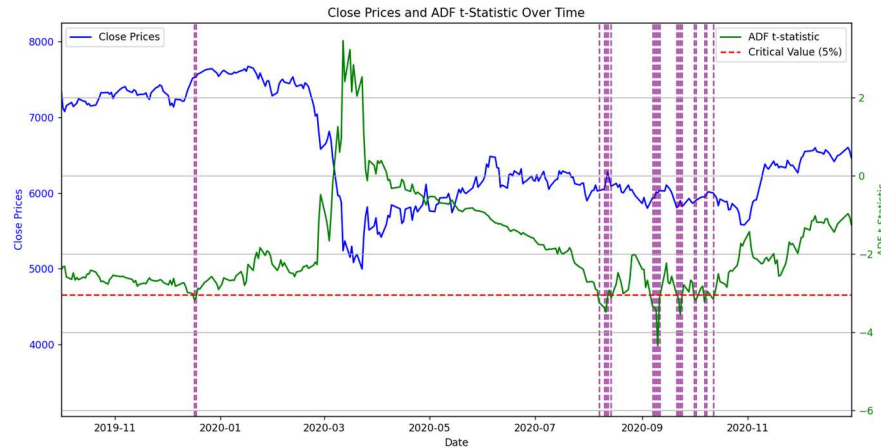
The above graph shows the predicted statistic plotted alongside FTSE returns. The statistic based on the analysed emotions was successfully able to predict upcoming negative returns in March 2020, which is when the UK government announced the lockdowns and businesses were impacted. Post the lockdown there are no significant signals indicating potential exponentially negative returns. The dates identified by the statistic were:

Start Date	03/02/2020	03/13/2020	11/09/2020
End Date	03/02/2020	03/13/2020	11/09/2020

Comparing the above result to verify with the recursive ADF statistic by Philips, Shapping and Jun we see that the recursive algorithm identifies explosiveness on the following date ranges:

Start Date	12/17/2019	12/18/2019	08/07/2020	08/10/2020	08/11/2020	08/12/2020
End Date	12/17/2019	12/18/2019	08/07/2020	08/10/2020	08/11/2020	08/12/2020
Start Date	08/14/2020	09/07/2020	09/21/2020	10/01/2020	10/07/2020	10/12/2020
End Date	08/14/2020	09/11/2020	09/24/2020	10/02/2020	10/08/2020	10/12/2020

Evaluated at the 5% significance it is seen that the explosiveness in the Auto Regressive model for prices occurs much after the bubble has already been burst and significant negative returns realized. Thus providing a much-delayed signal compared to the statistic that evaluates emotion explosiveness.



4.5 Model Parameters

Type	Emotion	PCA Lag in Days
Negative	anger	13
Neutral	anticipation	6
Negative	disgust	1
Negative	fear	7
Positive	joy	1
Negative	sadness	7
Neutral	surprise	5
Neutral	trust	1
Agreement Percentage = 51.987%		

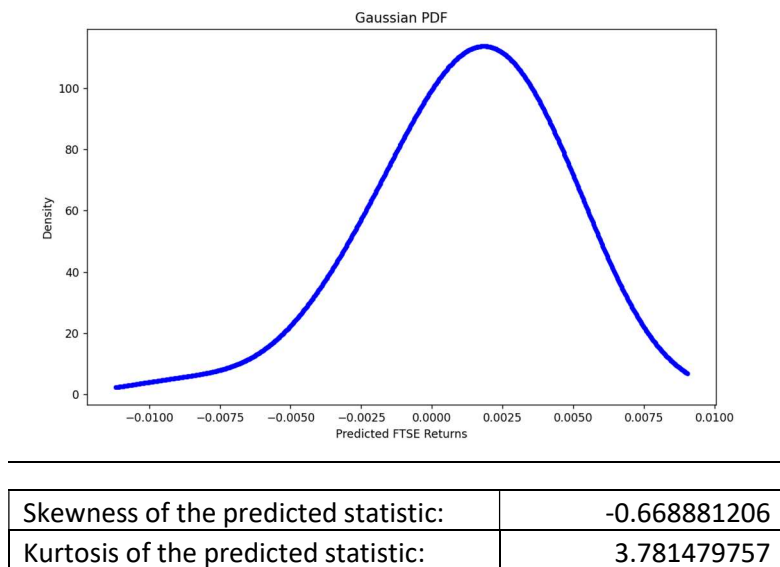
The above table indicates the lag period selected for the Principal Components for each emotion. This was done using the Mean Squared Error Loss function, which naturally gave the most optimum results. Due to the method of calculating the loss, the MSE function lays stronger emphasis on outliers and helps in closing the gap with those outliers when determining the parameters for a model. This played to an advantage in the given circumstances as the goal of the model was to predict significant negative moves in prices and capture the left extreme tail in the distribution of returns.

It was observed that negative emotions tend to start emerging in the financial literature much ahead in time before the actual response in the stock markets. Additionally, positive emotions seem to get immediately incorporated into the index prices as soon as it is discussed in the literature.

This shows that during a negative sentiment, investors prefer to hold on to their investments in anticipation of markets turning around, and only on confirmation of negative signals, they sell. Which also explains why “anticipation” and “surprise” have lag lengths close to those of the negative emotions.

Even though the purpose of the model is to predict extreme crisis events about to occur in the stock markets, an agreement percentage was calculated to verify day-to-day viability. This score was calculated to check, that if a binary classification is performed on the direction of the predicted statistic, into returns being positive or negative, and compared with the actual returns, the percentage of total number of times both lie on the same side. This statistic was calculated at 52.987%, being the number of times the prediction and true value lie on the same side.

4.6 Distribution of the Statistic



The above graph shows the Gaussian normal distribution of the predicted statistic. A kurtosis of 3.78 indicates an excess kurtosis of 0.78 which is close to the Gaussian normal distribution but has a higher probability of producing outliers compared to the normal distribution which works in favour of the model as this allows it to predict the extreme return cases more often.

A skewness of -0.67 indicates that the distribution is negatively skewed, which means that the left tail is fatter and longer than the right, increasing the probability of an extreme negative observation.

Thus, overall, a mildly higher than normal kurtosis with a negatively skewed tail makes the statistic more effective in predicting extreme negative returns in the market.

Section 5:
Conclusion

5.1 Primary Goal

Since the goal was to observe extreme behaviour in investors, a case study approach was followed. The model was trained for 9 years of textual data which was available through the Guardian API. Emotion scores were observed during four key events of crisis in the history of UK, namely; 2008 housing crisis which originated from the United States of America, 2011-12 Sovereign debt crisis that originated in Greece, significantly impacting the EU economies, the 2015-16 Brexit Referendum proposed by David Cameron which cut off UK from the European Union paving way for new trade treaties with the EU leading to uncertainty within the financial markets and lastly the Covid 19 crisis which forced businesses shut for a significant period due to a widespread pandemic.

The statistic works as a signalling mechanism, which creates a signal as soon as the value crosses below the two standard deviation mark indicating an upcoming crisis and exponentially negative returns. Based on this historical analysis, when comparing the results from 2008, 2011 and 2016 crisis with the Covid pandemic, it is seen that the method provides a much more effective and efficient signal in the former case studies.

This is mainly due to the nature of the crisis. The former three crisis are financial in nature in terms of their origin, however Covid being a medical crisis did not appear as significantly in the financial literature until March 2020, which is when businesses were halted, and the model captured a signal for probable negative returns. This suggests that the model can analyse literature and capture only those signals which are financial in nature. Since the first mention of Covid was made in the Guardian under their world news section in an article named “Chinese-pneumonia-outbreak-may-be-caused-by-new-virus”¹² in January 2020 which was not captured by the model.

5.2 PSY Validation

The output generated was compared with the recursive ADF test developed by Peter C.B. Phillips, Shuping Shi, and Jun Yu (2015) to verify presence of bubbles through presence of mild explosiveness in an Auto Regressive time-series.

The ADF test confirms the accuracy of bubbles through the method developed. However, the PSY ADF statistic produces a delayed signal with respect to the textual model in use. Negative emotions were observed seven to eight days prior to the index prices declining, indicating that investors hold on to investments in anticipation of the markets reversing and only upon negative returns plunging below their loss bearing capacity, an exit trade is made, despite strong criticism towards market conditions in the financial literature.

5.3 Applications

The above study offers significant applications across various domains of financial risk management. In addition to its utility in Value at Risk (VaR) modelling, where it effectively captures the left extreme endpoints in the tail of the distribution, the method serves as an early warning system, allowing investors to anticipate market downturns and optimize strategies accordingly.

Furthermore, it can be instrumental in monitoring systemic risk across financial markets through analysis of textual data collected from a wide range of sources—such as news articles, financial

¹² <https://www.theguardian.com/world/2020/jan/07/chinese-pneumonia-outbreak-may-be-caused-by-new-virus>

reports, and social media. Banks and Financial Institutions can detect early signs of stress within the economy, thus equipping them with the necessary tools to undertake pre-emptive measures to mitigate significant financial losses.

Additionally, Companies may be able to leverage this approach within corporate governance to anticipate potential risks and opportunities that may impact their strategies and future growth plans. By incorporating signals from textual data, businesses can make more informed decisions with respect to various corporate actions such as mergers or acquisitions.

5.4 Limitations

Even though the above-mentioned approach is effective in predicting the extreme left tail values, it is relatively inefficient when observing the right tail. This is because the model relies on high volume of terms for an emotion to be able to capture the signal, due to the challenges faced in preprocessing of text, which tends to leave relatively higher volume of noise than when dealing with numerical data, and more often than not, humans tend to have a “negativity bias” which is shown by Roy F. Baumeister, Ellen Bratslavsky, Catrin Finkenauer, and Kathleen D. Vohs (2001) where they conclude that negative experiences and interactions are often more potent and have a longer-lasting effect than positive ones and thus there is higher volume of data for negative events, than positive.

The model can be optimized further to capture positive signals better and predict the right tail. However, firstly, this would require the dictionary to be near perfect with respect to the terms that appear in literature signalling positive emotions, coupled with a weighted tf-idf technique which gives higher weight to positive terms and secondly, there would still be a strong challenge in being able to capture the signals before time, as seen in the best lag generated through lowest MSE in the multilinear regression, the positive emotions have a mere one day lag, which too could be late at times since positive news tends to get incorporated into prices much faster.

Another downside of using such a comprehensive approach instead of a mere positive/ negative word classification to do a binary prediction for the day is that due to the large volume of signals that the model aims to capture, it also captures more noise compared to the binary classification models, which may not work as effectively when trying to use this approach for predicting day to day trading directions. This method has been able to predict the direction of a trading day with roughly 52% accuracy while the binary approach has shown better results, such as Luis Villamil, Ryan Bausback, Shaeke Salman, Ting L. Liu, Conrad Horn, Xiuwen Liu (2023) which reported an accuracy of 65%.

5.5 Further Research

As grounds for additional research, the PSY test can be performed on the principal components for each of the emotion's scores individually through an Autoregressive model. This could help check for explosiveness within each emotion individually. Further, comparing this explosiveness with time periods of significant returns, and extracting only those negative emotions which show simultaneous explosiveness and then using them for the purpose of calculating the statistic. This would help reduce the dimensions of the overall problem and predicting bubbles further ahead in time. Moreover, this could also help improve the binary classification of positive and negative trading days, where the aim would be to only capture the negativity within the literature, and days of low or no negativity may be classified as a positive trading day.

Additionally, following the approach of Samuel P. Fraiburger (2016), economic variables available daily can be combined with the textual approach and incorporated as inputs within the multilinear regression. This could help create a balance between the comprehensive and binary approach with improved accuracy in predicting the direction for any given trading day well ahead in time.

In conclusion the model developed through this study is successfully able to achieve its primary objective alongside an added advantage.

The model successfully predicts major upcoming financial crisis accurately providing a signal 7 to 8 days ahead in time versus the existing PSY test which tends to give a relatively delayed signal at times, after the index has already plunged. With the fast-changing environment and use of technology in financial markets, being able to generate timely signals has become more crucial to be able to take advantage of the full swing in prices.

Additionally, emotions prevailing within the financial literature can be also used to predict the direction of the index on a given trading day, with 52% accuracy, sufficient to ensure overall profitability when using the model.

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